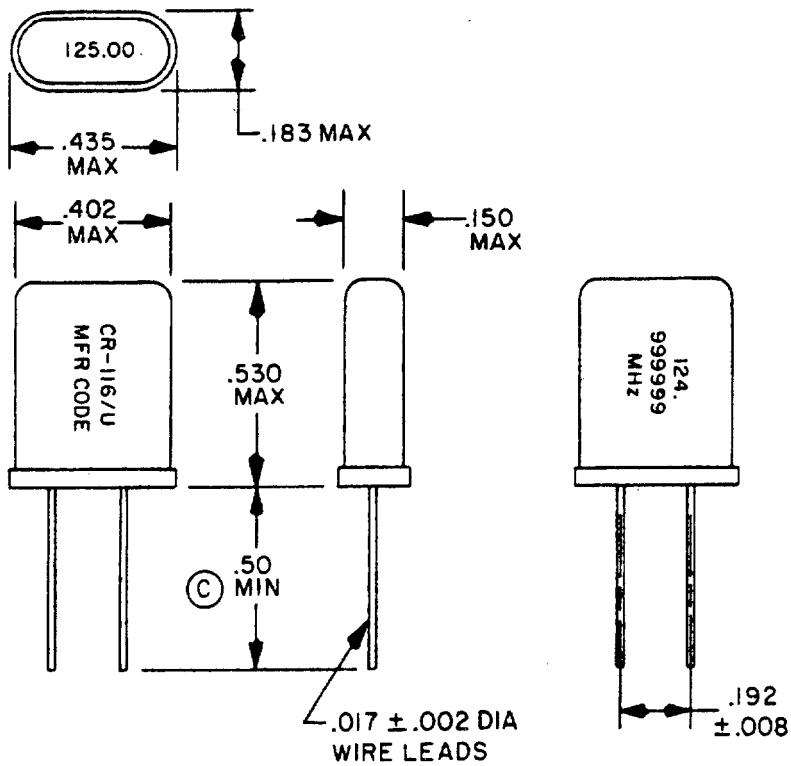
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-116/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; noncontrolled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-116/U.

© denotes changes

FSC 5955

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© TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	50 to 125 MHz, inclusive
Capacitance, shunt	4.5 pF, maximum
Mode of oscillation	Fifth mechanical overtone
Government reference standard test sets	TS-683/TSM and AN/TSM-15
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Resonance	Series
Operating temperature range (noncontrolled)	-55°C to +105°C, inclusive
Frequency tolerance	±25 ppm
Equivalent resistance	60 ohms, maximum
Unwanted modes 1/	Method II
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Unwanted modes: The minimum unwanted mode effective resistance shall be at least three times that of the main mode, or 120 ohms, whichever is greater.

2/ Method 204 of MIL-STD-202, test condition A.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activity:
Air Force - 17

Agent:
DLA - ES

Preparing activity:
Army - ER

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